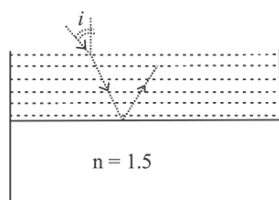


JEE Mains 2019 Chapter wise Question Bank

Wave Optics - Questions

Q1

Consider a tank made of glass (refractive index 1.5) with a thick bottom. It is filled with a liquid of refractive index μ . A student finds that, irrespective of what the incident angle i (see figure) is for a beam of light entering the liquid, the light reflected from the liquid glass interface is never completely polarized. For this to happen, the minimum value of μ is:



- (1) $\sqrt{\frac{5}{3}}$ (2) $\frac{3}{\sqrt{5}}$
 (3) $\frac{5}{\sqrt{3}}$ (4) $\frac{4}{3}$

9 Jan Morning

Q2

In a Young's double slit experiment, the slits are placed 0.320 mm apart. Light of wavelength $\lambda = 500$ nm is incident on the slits. The total number of bright fringes that are observed in the angular range $-30^\circ \leq \theta \leq 30^\circ$ is

- (1) 640 (2) 320
 (3) 321 (4) 641

9 Jan Evening

Q3

In a Young's double slit experiment with slit separation

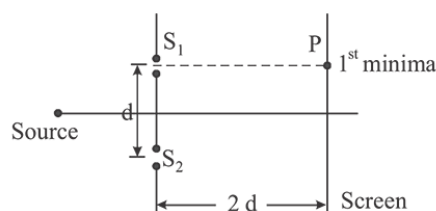
0.1 mm, one observes a bright fringe at angle $\frac{1}{40}$ rad by using light of wavelength λ_1 . When the light of wavelength λ_2 is used a bright fringe is seen at the same angle in the same set up. Given that λ_1 and λ_2 are in visible range (380 nm to 740 nm), their values are:

- (1) 625 nm, 500 nm (2) 380 nm, 525 nm
 (3) 380 nm, 500 nm (4) 400 nm, 500 nm

10 Jan Morning

Q4

Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1)?



- (1) $\frac{\lambda}{2(\sqrt{5}-2)}$ (2) $\frac{\lambda}{(\sqrt{5}-2)}$
 (3) $\frac{\lambda}{2(5-\sqrt{2})}$ (4) $\frac{\lambda}{(5-\sqrt{2})}$

10 Jan Evening

Q5

In a Young's double slit experiment, the path difference, at a certain point on the screen, between two interfering waves is $\frac{1}{8}$ th of wavelength. The ratio of the intensity at this point to that at the centre of a bright fringe is close to:

- (1) 0.74 (2) 0.85
 (3) 0.94 (4) 0.80

11 Jan Morning

Q6

In a double-slit experiment, green light ($5303\text{\AA}$) falls on a double slit having a separation of $19.44\text{ }\mu\text{m}$ and a width of $4.05\text{ }\mu\text{m}$. The number of bright fringes between the first and the second diffraction minima is:

- (1) 10 (2) 05
 (3) 04 (4) 09

11 Jan Evening

Wave Optics

Q7

In an interference experiment the ratio of amplitudes of coherent waves is $\frac{a_1}{a_2} = \frac{1}{3}$. The ratio of maximum and minimum intensities of fringes will be :

- (1) 2 (2) 18 (3) 4 (4) 9

8 April Morning

Q8

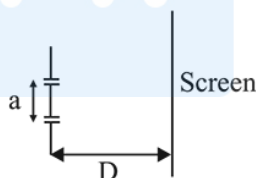
Calculate the limit of resolution of a telescope objective having a diameter of 200 cm, if it has to detect light of wavelength 500 nm coming from a star.

- (1) 305×10^{-9} radian (2) 610×10^{-9} radian
(3) 152.5×10^{-9} radian (4) 457.5×10^{-9} radian

8 April Evening

Q9

The figure shows a Young's double slit experimental setup. It is observed that when a thin transparent sheet of thickness t and refractive index μ is put in front of one of the slits, the central maximum gets shifted by a distance equal to n fringe widths. If the wavelength of light used is λ , t will be:



- (1) $\frac{2nD\lambda}{a(\mu-1)}$ (2) $\frac{nD\lambda}{a(\mu-1)}$
(3) $\frac{D\lambda}{a(\mu-1)}$ (4) $\frac{2D\lambda}{a(\mu-1)}$

9 April Morning

Q10

Diameter of the objective lens of a telescope is 250 cm. For light of wavelength 600 nm. Coming from a distant object, the limit of resolution of the telescope is close to:

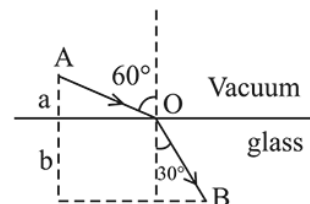
- (1) 1.5×10^{-7} rad (2) 2.0×10^{-7} rad
(3) 3.0×10^{-7} rad (4) 4.5×10^{-7} rad

9 April Evening

Q11

JEE Mains 2019 Chapter wise Question Bank

A ray of light AO in vacuum is incident on a glass slab at angle 60° and refracted at angle 30° along OB as shown in the figure. The optical path length of light ray from A to B is :



- (1) $\frac{2\sqrt{3}}{a} + 2b$ (2) $2a + \frac{2b}{3}$
(3) $2a + \frac{2b}{\sqrt{3}}$ (4) $2a + 2b$

10 April Morning

Q12

In a Young's double slit experiment, the ratio of the slit width is 4 : 1. The ratio of the intensity of maxima to minima, close to the central fringe on the screen, will be :

- (1) 25 : 9 (2) 9 : 1
(3) 4 : 1 (4) $(\sqrt{3} + 1)^4 : 16$

10 April Evening

Q13

The value of numerical aperture of the objective lens of a microscope is 1.25. If light of wavelength 5000 Å is used, the minimum separation between two points, to be seen as distinct, will be :

- (1) 0.24 μm (2) 0.38 μm (3) 0.12 μm (4) 0.48 μm

12 April Morning

Q14

In a double slit experiment, when a thin film of thickness t having refractive index μ is introduced in front of one of the slits, the maximum at the centre of the fringe pattern shifts by one fringe width. The value of t is (λ is the wavelength of the light used) :

- (1) $\frac{2\lambda}{(\mu-1)}$ (2) $\frac{\lambda}{2(\mu-1)}$
(3) $\frac{\lambda}{(\mu-1)}$ (4) $\frac{\lambda}{(2\mu-1)}$

12 April Morning

Q15

A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1 . The pass axis of P_2 is inclined at 60° to the pass axis of P_3 . When a beam of unpolarized light of intensity I_0 is incident on P_1 , the intensity of light transmitted by the three polarizers is I .

The ratio (I_0/I) equals (nearly) :

- (1) 5.33 (2) 16.00 (3) 10.67 (4) 1.80

12 April Evening



mathongo

JEE Mains 2019 Chapter wise Question Bank

Wave Optics - Questions

Q1

- (2) According to Brewster's law, refractive index of material (μ) is equal to tangent of polarising angle

$$\therefore \tan i_b = \mu = \frac{1.5}{\mu}$$

$$\frac{1}{\mu} < \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}} \quad (\because \sin i_c < \sin i_b)$$

$$\therefore \sin i_b = \frac{1.5}{\sqrt{\mu^2 + (1.5)^2}}$$

$$\text{or, } \sqrt{\mu^2 + (1.5)^2} < 1.5 \times \mu$$

$$\Rightarrow \mu^2 + (1.5)^2 < (\mu \times 1.5)^2$$

$$\Rightarrow \mu < \frac{3}{\sqrt{5}} \text{ i.e. minimum value of } \mu \text{ should}$$

$$\text{be } \frac{3}{\sqrt{5}}$$

9 Jan Morning

Q2

- (4) For 'n' number of maximas

$$d \sin \theta = n\lambda$$

$$0.32 \times 10^{-3} \sin 30^\circ = n \times 500 \times 10^{-9}$$

$$\therefore n = \frac{0.32 \times 10^{-3}}{500 \times 10^{-9}} \times \frac{1}{2} = 320$$

Hence total no. of maximas observed in angular range -30°

$$\leq \theta \leq 30^\circ$$

$$= 320 + 1 + 320 = 641$$

9 Jan Evening

Q3

- (1) Path difference = $d \sin \theta \approx d\theta$

$$= 0.1 \times \frac{1}{40} \text{ mm} = 2500 \text{ nm}$$

For bright fringe, path difference must be integral multiple of λ .

$$\therefore 2500 = n\lambda_1 = m\lambda_2$$

$$\therefore \lambda_1 = 625 \text{ (for } n = 4), \lambda_2 = 500 \text{ (for } m = 5)$$

10 Jan Morning

Q4

- (1) Here, $x_1 = 2d$ and $x_2 = \sqrt{5}d$

$$\text{For, first minima, } \Delta x = \frac{\lambda}{2}$$

$$\therefore \Delta x = x_2 - x_1 = \sqrt{5}d - 2d = \frac{\lambda}{2}$$

$$\Rightarrow d = \frac{\lambda}{2(\sqrt{5} - 2)}$$

10 Jan Evening

Q5

- (2) Given, path difference, $\Delta x = \frac{\lambda}{8}$

Phase difference ($\Delta\phi$) is given by

$$\Delta\phi = \frac{2\pi}{\lambda}(\Delta x)$$

$$\Delta\phi = \frac{(2\pi)\lambda}{\lambda} \frac{1}{8} = \frac{\pi}{4}$$

For two sources in different phases,

$$I = I_0 \cos^2\left(\frac{\pi}{8}\right)$$

$$\frac{I}{I_0} = \cos^2\left(\frac{\pi}{8}\right)$$

$$= \frac{1 + \cos \frac{\pi}{4}}{2} = \frac{1 + \frac{1}{\sqrt{2}}}{2} = 0.85$$

11 Jan Morning

Q6

- (2)

11 Jan Evening

Wave Optics

JEE Mains 2019 Chapter wise Question Bank

Q7

(3) Given amplitude ratio of waves is $\frac{a_1}{a_2} = \frac{3}{1}$

so, $\frac{I_{\max}}{I_{\min}} = \left(\frac{a_2 + a_1}{a_2 - a_1} \right)^2$

$$= \left(\frac{\frac{a_2}{a_1} + 1}{\frac{a_2}{a_1} - 1} \right)^2 = \left(\frac{3+1}{3-1} \right)^2 = \left(\frac{4}{2} \right)^2 = \frac{4}{1} = 4$$

8 April Morning

Q8

(1) $\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 500 \times 10^{-9}}{2} = 305 \times 10^{-9} \text{ rad.}$

8 April Evening

Q9

(Bonus) Shift = $n\beta$ (given)

$$\therefore D \frac{(\mu-1)t}{a} = \frac{n\lambda D}{a} \left[\because \text{Shift} = \frac{D(\mu-1)t}{a} \right]$$

$$\text{or } t = \frac{n\lambda}{(\mu-1)}$$

9 April Morning

Q10

(3) $\theta = \frac{1.22\lambda}{d} = \frac{1.22 \times 600 \times 10^{-9}}{250 \times 10^{-2}}$

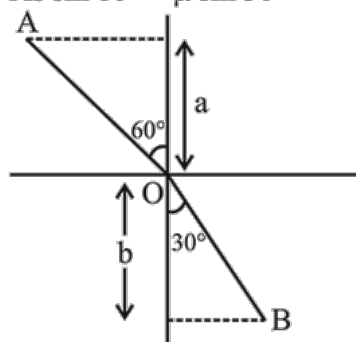
$$= 3.0 \times 10^{-7} \text{ rad}$$

9 April Evening

Q11

(4) From the given figure

As $\sin 60^\circ = \mu \sin 30^\circ$



$$\Rightarrow \mu = \frac{\sin 60^\circ}{\sin 30^\circ} = \sqrt{3}$$

$$\frac{a}{AO} = \cos 60^\circ \Rightarrow AO = 2a$$

$$\frac{b}{BO} = \cos 30^\circ \Rightarrow BO = \frac{2b}{\sqrt{3}}$$

Optical path length = $AO + \mu BO$

$$= 2a + (\sqrt{3}) \frac{2b}{\sqrt{3}} = 2a + 2b$$

10 April Morning

Q12

(2) Let I_1, I_2 are intensity of light coming from two slits.

As, $I \propto W$

$$\therefore \frac{I_1}{I_2} = \frac{4}{1}$$

$$I_1 = 4 I_0$$

$$I_2 = I_0$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2} = \left(\frac{9}{1} \right)$$

$$= \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right)^2$$

$$= \left(\frac{\sqrt{4} + 1}{\sqrt{4} - 1} \right)^2 = \frac{9}{1}$$

10 April Evening

Q13

$$(1) \ x = \frac{1.22\lambda}{2\mu\sin\theta}$$

$$= \frac{1.22 \times 5000}{2 \times 1.25} = 0.24 \mu\text{m}$$

12 April Morning

Q14

(3) Given, $\Delta = \beta$

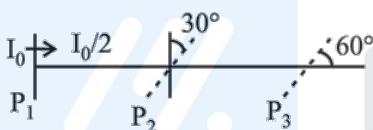
$$\text{or } \frac{D(\mu-1)t}{d} = \frac{D\lambda}{d}$$

$$\therefore t = \frac{\lambda}{(\mu-1)}$$

12 April Morning

Q15

$$(3) \ I = \left(\frac{I_0}{2} \right) \cos^2 30^\circ \cos^2 60^\circ$$



$$= \frac{I_0}{2} \times \frac{3}{4} \times \frac{1}{4}$$

$$\therefore \frac{I_0}{I} = \frac{32}{3} = 10.67$$

12 April Evening